When Does Pooling Make Intersectional Fairness Audits Falsely Reassuring?

ICT 5143: Algorithms Fairness and Graphical Models Laboratory
Problem 8 (Reliable intersectional fairness audits); Team 5
Sumedh Navuda (261090830007); Aldino (261090830006)
sumedh.mitmpl2026@learner.manipal.edu; aldino.mitmpl2026@learner.manipal.edu
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Abstract

We ask when Bayesian pooling makes an intersectional equal-opportunity audit report "gap ≤ 0.10" for an outreach rule that is unfair. A discrete Bayesian network ranks ACSPublicCoverage adults (California 2018), and the audits estimate race×sex×disability TPRs. Pooled audits hid isolated small-group deficits, and all 116 California and 27 Texas validation-time tolerance claims failed on test. Calibrating the audit on the target population's group sizes and certifying on held-out households raised the audit score on fresh Texas simulations by +0.279 (95% CI $[0.235, 0.322]$). Limitation: on California it only ties the best pooled audit.

1 Question, context and contribution

An agency that can contact 30% of a list sends *hypothetical* information outreach ($D = 1$) to the people ranked most likely to have no public coverage ($Y = 1$, the ACS record; this is neither uninsurance nor eligibility). Those harmed are people without coverage who are never contacted, often in small intersections such as disabled Black women. **Research question:** when does hierarchical pooling stabilise intersectional TPR estimates, and when does it mask a harmful small-group disparity, so that a tolerance claim $\Delta\text{EOpp} \leq \epsilon$ is false?

Contribution: pooled audit criteria used to choose a group-threshold rule on validation data make tolerance claims that fail on test in more than half of cases (H1; refuted if at most half fail on Texas). Design-matched calibration with held-out certification raises the audit score J by at least 0.10 over every established audit (H2; refuted if not).

Closest work: hierarchical Bayesian intersectional fairness [1] and structured regression for small subgroups [2] pool sparse groups, and Bayesian [3], variance-corrected [4] and simultaneous-bootstrap [5] audits quantify uncertainty. None reports how often a pooled audit falsely certifies a gap tolerance when one small intersection is harmed, or when the audit is used to choose the rule. That marginal fairness can hide subgroup harm [6] and that fairness constraints can fail out of sample [7] is known. Our `struct` audit is a restricted version of [1], not an exact reproduction, and the contamination prior is classical [8]. We therefore claim an evaluation and an audit protocol, not a new prior or metric. The full 19-paper comparison is in the repository.

2 Data, graphical model and fairness method

Data and estimand: ACS PUMS 2018 1-year via Folktables ACSPublicCoverage [9] (AGEP < 65, PINCP $\leq 30{,}000$): 138,554 Californians in 85,182 households, $P(Y{=}1) = 0.631$, with $Y = 1 - \mathbf{1}[\text{PUBCOV} = 1]$. Protected attributes are recorded race (9 codes), sex and disability, giving 36 intersections; only 21 of 35 observed test groups have ≥ 30 positives. Splits: 60/20/20 *by household*, 5 seeds; estimand: the unweighted sample; replication: Texas 2018 (98,928).

Graph and inference (Figure 1): the predictor is an augmented naive-Bayes network (associational, not causal) with Dirichlet $\alpha = 1$ CPTs, queried by exact variable elimination. The audit is a second graphical model over group counts.

$$p(y, \mathbf{x}) = p(y) \textstyle\prod_j p(x_j \mid \text{pa}(x_j)), \qquad s(\mathbf{x}) = P(Y{=}1 \mid \mathbf{x}) \text{ (variable elimination);}$$
$$k_g \sim \text{Bin}(n_g, \theta_g), \qquad \theta_g \sim \pi\,\text{Beta}(a, b) + (1 - \pi)\,\text{Beta}\big(\mu c, (1 - \mu)c\big), \quad \pi \leq \pi_{\max}.$$

The comparators replace this prior with: none (raw); independent Beta(1,1); complete pooling; exchangeable Beta(a, b) by empirical Bayes or a full-Bayes grid; and `struct`, which is $\text{logit}\,\theta_g = \beta_0 + \beta_{\text{race}} + \beta_{\text{sex}} + \beta_{\text{dis}} + u_g$ with $u_g \sim N(0, \tau^2)$, fitted by quadrature.

Method and fairness: the primary criterion is $\Delta\text{EOpp} = \max_g \text{TPR}_g - \min_g \text{TPR}_g$ with $\epsilon = 0.10$,

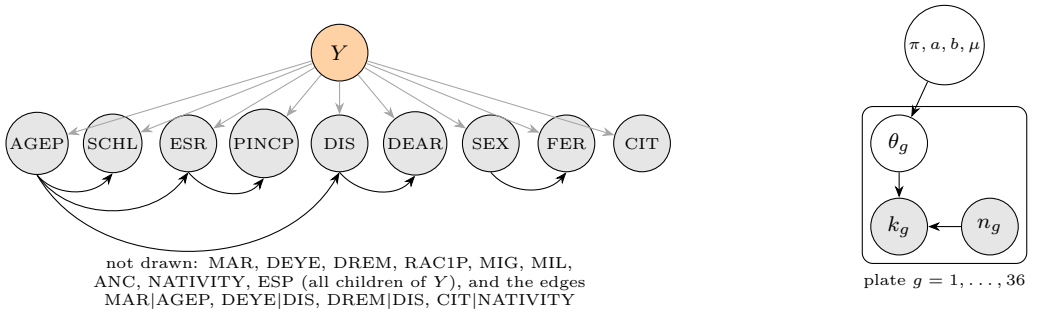


Figure 1: Left: predictive BN; grey arrows $Y \to$ feature, black arrows the feature edges; shaded nodes are observed. Right: audit model, with k_g of the n_g positives in group g selected.

because the harm is missing people who need outreach. Diagnostics: ΔDP, ΔEOdds, selection ratio, worst-group recall, group calibration. Utility is recall at 30% capacity, with at most 0.02 loss. A rule family $R(\lambda)$ blends global and group thresholds, so A is used at prediction time (by the scores, and by the thresholds for $\lambda > 0$). An audit certifies when its plug-in or upper-q posterior gap is $\leq \epsilon$. **Proposed:** choose $\pi_{\max}$, the slab and the rule by maximising $J = \text{ACC} - \overline{\text{FTS}}$ on semi-synthetic worlds built on the *target* state's validation group sizes. Then choose λ on one half of the validation households and certify on the other half.

3 Evaluation protocol

Baselines: logistic regression and gradient boosting (predictive); the BN without uncertainty-aware selection (graphical); raw, independent, exchangeable, complete and `struct` audits (fairness). All audits see the same counts and all rules the same validation-matched capacity. Training fits edges and CPTs, validation fits λ and thresholds, and test only evaluates frozen rules. Audits are scored on simulated worlds with known truth: fair worlds, and worlds with a 0.20 TPR deficit in an isolated cell, a margin-balanced interaction or a disability main effect, at full and half support; calibration and evaluation use different seeds. Real comparisons use five household splits and a household-cluster Poisson bootstrap (B = 1000) with Holm correction over the two preregistered comparisons; ΔJ uses a paired bootstrap over simulated worlds (B = 2000). Exact VE matches pgmpy VE, clique-tree propagation and enumeration to 3.3×10^{-16} (tolerance 10^{-9}). All five mandatory experiments, the ablations and the robustness tests (priors, prevalence, ϵ, thinning, second state) were run.

4 Results and interpretation

Audit procedure (fresh Texas worlds, 200 per cell)	ACC (fair, ↑)	mean FTS (↓)	J (↑)	90% gap coverage
Closest baseline: full-Bayes beta-binomial, plug-in	1.000	0.615	0.385	0.174
Hierarchical EB, upper 0.9	0.985	0.625	0.360	0.013
`struct` (Foulds-style), plug-in	0.108	0.073	0.034	0.037
Ablation: no design matching	0.218	0.028	0.189	0.506
Proposed: design-matched contamination audit	0.832	0.168	**0.664**	0.615
Real CA+TX, 10 splits	claims / false claims		test recall	test ΔEOpp
Ablation: same-data certification	10 / 10		0.365	0.302
Proposed: held-out certification	0 / 0		0.365	0.309

Table note: ACC/FTS = share of fair/unfair worlds certified ($\epsilon = 0.10$; FTS averaged over 3 deficit types $\times$ 2 support levels). Paired ΔJ, proposed minus baseline: +0.279 [0.235, 0.322] vs full Bayes, +0.475 [0.422, 0.525] vs the ablation, −0.004 [−0.057, 0.046] on the CA design (tie). Shared predictor quality: BN AUROC 0.729 ± 0.004, log loss 0.577, Brier 0.196 (GBT 0.768, 0.541, 0.182).

Main finding: pooling helps when a deficit sits in large groups or a main effect and masks it when the deficit is isolated. For fixed BN decisions the gap is 0.162 by sex and 0.746 over the 21 supported intersections; this disparity is structural, so every audit gives $P(\Delta\text{EOpp} > 0.10) = 1$. On synthetic

worlds EB upper-0.9 falsely certified 98% of worlds with one 20-positive group 0.20 below the rest. All 116 CA and 27/27 TX pooled validation claims failed on test (gap 0.110–0.888): **H1 supported**. The proposed audit beats every established audit and the ablation by > 0.10 in J, and held-out certification withheld all 10 false claims at unchanged recall (**H2a–c supported**).

Mechanism and failure: without design matching, fair-world acceptance drops from 0.83 (invented sizes) to 0.22 (Texas); 90% of false alarms come from groups with < 10 positives. On California the method trades lower FTS (0.348 vs 0.666) for lower ACC (0.678), so J ties. No rule is fair on test (min gap 0.227); the preregistered real-gap comparison is inconclusive (-0.052 [-0.194, 0.051], Holm $p = 0.736$); `struct` under-covers (0.665 at nominal 0.90). Groups with < 30 positives are exploratory.

5 Limitations, ethics and conclusion

Real-data facts are the counts, gaps and failed claims; the J gains are simulation evidence from one generator family on real group sizes. Race and disability are survey constructs, and the label is not need, so the model can misdirect resources. Zero held-out claims shows abstention, not sensitivity to rules that really are fair. No stakeholders set ϵ or the capacity. **Conclusion:** do not trust a pooled "gap $\leq \epsilon$" for sparse intersections unless the audit was calibrated on that design and certified on households not used to choose the rule. Next experiment: an unseen generator family, plus a constructed fair rule to measure held-out sensitivity.

Reproducibility and individual contributions

Reproduce: `jupyter nbconvert --to notebook --execute --inplace Problem8_Intersectional_Fairness.ipynb` (about 30 min on 1 CPU core; Python 3.12, `requirements-lock.txt`), then `python writeup/make_tables.py` for the ΔJ intervals. Config: notebook §0 constants, mirrored in `configs/frozen_settings.yaml`.

Sumedh Navuda: question, preregistration, literature, data and splits, BN and exact-inference checks, Experiment 1, Texas H1 test; cross-checked the audit posteriors, Holm and J intervals.

Aldino: audit models and coverage checks, simulations and calibration, threshold selection, bootstrap, the design-matched/held-out procedure (H2); cross-checked splits, label reversal and inference agreement.

Evidence: `docs/requirements_matrix.md`, `docs/contributions.md`, `results/`.

References

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